



GOVERNMENT POLYTECHNIC, SIRA

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BIODIESEL

A SUSTAINABLE ALTERNATIVE FUEL

ENTREPRENEURSHIP IN BIODIESEL

From Opportunity to Sustainable Impact



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RENEWABLE
ENERGY



ECO-FRIENDLY



SUSTAINABLE
FUTURE



CLEANER
PLANET

Introduction to Biodiesel



- Biodiesel is a clean and renewable fuel made from vegetable oils, animal fats, or used cooking oil. It is an eco-friendly alternative to petroleum diesel and can be used in most diesel engines.
- **What is Biodiesel?**
- **diesel** is a renewable fuel produced from natural oils and fats through a process called **transesterification**. It can be used alone or mixed with diesel fuel.
- **Why is Biodiesel Important?**
- Reduces air pollution.
- Renewable and eco-friendly.
- Biodegradable and non-toxic.
- Reduces dependence on fossil fuels.
- Can be used in existing diesel engines with little or no modification.



Problem Statement



- **Waste Cooking Oil Disposal:** Improper disposal of used cooking oil causes waste management problems and blocks drains.
- **Water Pollution:** Waste cooking oil pollutes rivers, lakes, and groundwater, harming aquatic life.
- **Environmental Impact:** Improper disposal and the use of fossil diesel increase pollution and greenhouse gas emissions.
- **Need for a Solution:** Converting waste cooking oil into biodiesel reduces waste, protects the environment, and provides a renewable fuel source.



Objectives

- **Recycle Waste Cooking Oil:** Convert used cooking oil into a useful renewable fuel instead of disposing of it.
- **Produce Eco-Friendly Fuel:** Manufacture biodiesel as a clean and sustainable alternative to petroleum diesel.
- **Reduce Pollution:** Minimize air, water, and soil pollution by recycling waste oil and reducing harmful emissions.



Why Biodiesel from Cooking Oil?

- **Reuses waste oil** instead of throwing it away.
- **Reduces water pollution** caused by improper oil disposal.
- **Produces eco-friendly fuel** from a waste material.
- **Reduces dependence on petroleum diesel.**
- **Helps reduce pollution** and environmental damage.
- **Low-cost raw material** because used cooking oil is easily available.



Raw Materials Required

- **Waste cooking oil** – main raw material
- **Methanol** – used to convert oil into biodiesel
- **Sodium hydroxide (NaOH)** – used as a catalyst
- **Water** – used for washing and cleaning the biodiesel
- **Used cooking oil filter** – removes food particles and impurities

